



**Università
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High Performance Computing Report CUDA

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1 Architectures

For this project, I used two workstations: my personal laptop and the Google Colab environment

HP Envy 17-ae103nl (Laptop)

CPU

- **Model:** Intel Core i7-6500U @ 2.50 GHz
- **Architecture:** x86_64
- **Physical cores:** 2
- **Logical threads:** 4 (Hyper-Threading)

GPU

- **Model:** NVIDIA GeForce GTX 950M
- **CUDA Compute Capability:** 5.0
- **Driver version:** 572.16
- **CUDA Version:** 12.8 (*CUDA Toolkit 12.x*)

Google Colab

CPU

- **Model:** Intel(R) Xeon(R) CPU @ 2.20GHz
- **Architecture:** x86_64
- **Physical cores:** 1
- **Logical threads:** 2 (Hyper-Threading)
- **Socket(s):** 1

GPU

- **Model:** NVIDIA Tesla T4
- **CUDA Compute Capability:** 7.5
- **Driver version:** 550.54.15
- **CUDA Version:** 12.8 (*CUDA Toolkit 12.x*)

2 Hotspot Analysis

3 Vectorization

4 Parallelization

5 Conclusions

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